

NCT07562074: Transcranial Magnetic Stimulation for Depression in Multiple Sclerosis

Gender: ALL

Age: 18 Years - 65 Years

Criteria:

Inclusion Criteria:

1. 18-65 years of age, inclusive.
2. Males, females, and non-binary.
3. Clinical diagnosis of relapsing-remitting or primary-progressive multiple sclerosis according to the revised McDonald criteria and as determined by the study neurologist and medical records.
4. Expanded Disability Status Scale score ≤ 7 .
5. Fatigue Severity Scale ≥ 4 .
6. Moderate to severe major depressive disorder as defined by a Hamilton Depression Rating Scale-17 item score ≥ 18 .
7. Did not benefit adequately from ≥ 1 antidepressant or course of psychotherapy.
8. Stable immunotherapy for at least 3 months and medications for at least 4 weeks.
9. Have no contraindications to TMS or magnetic resonance imaging.
10. Ability to provide written consent obtained from study subject or subject's legal representative and ability for study subject to comply with the requirements of the study.
11. Are able to adhere to the treatment schedule
12. Pass the TMS adult safety screening (TASS) questionnaire

Exclusion Criteria:

1. Presence of any disease, medical condition, or physical condition that, in the opinion of the study investigator, study psychiatrist, or study neurologist, may compromise interfere, limit, affect, or reduce the study subject's ability to complete the study.
2. Presence of any disease, medical condition, or physical condition that, in the opinion of the study investigator, study psychiatrist, or study neurologist, may adversely impact the safety of the study subject or the integrity of the data.
3. History of seizures.
4. Any cranial metal implants (excluding ≤ 1 mm thick epicranial titanium skull plates and dental fillings) or medical devices (i.e. cardiac pacemaker, deep brain stimulator, medication infusion pump, cochlear implant).
5. Previous surgeries opening the skull leaving skull defects capable of allowing the insertion of a cylinder with a radius ≥ 5 mm.
6. Any diagnosis of a psychiatric diagnosis determined to be primary.
7. Are at a significant risk of harm to themselves or others.
8. Have a substance or alcohol use disorder within the last three months.
9. If participating in psychotherapy, must have been in stable treatment for at least three months prior to entry into the study, with no anticipation of change in the frequency of therapeutic sessions, or the therapeutic focus over the duration of the study.
10. Are currently pregnant, breast feeding, or plan to become pregnant over the duration of the study.
11. Active suicidal ideation as defined by a score of 4 $\geq$ on item 10 of MADRS
12. Have failed a course of ECT in the current episode. Previous ECT treatment outside of the current episode does not influence inclusion.
13. Have any significant neurological disorder or insult including, but not limited to: any condition likely to be associated with increased intracranial pressure, space occupying brain lesion, any history of epilepsy, cerebral aneurysm, Parkinson's disease, Huntington's chorea, significant head trauma with loss of consciousness for greater than or equal to 5 minutes
14. Have concomitant major unstable medical illness